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Risk factors affecting polygenic score performance across diverse cohorts

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Abstract

Apart from ancestry, personal or environmental covariates may contribute to differences in polygenic score (PGS) performance. We analyzed effects of covariate stratification and interaction on body mass index (BMI) PGS (PGS_{BMI}) across four cohorts of European (N=491,111) and African (N=21,612) ancestry. Stratifying on binary covariates and quintiles for continuous covariates, 18/62 covariates had significant and replicable R^2 differences among strata. Covariates with the largest differences included age, sex, blood lipids, physical activity, and alcohol consumption, with R^2 being nearly double between best and worst performing quintiles for certain covariates. 28 covariates had significant PGS_{BMI}-covariate interaction effects, modifying PGS_{BMI} effects by nearly 20% per standard deviation change. We observed overlap with covariates that had significant R^2 differences between strata and interaction effects – across all covariates, their main effects on BMI were correlated with maximum R^2 differences and interaction effects (0.56 and 0.58, respectively), suggesting high-PGS-score individuals have highest R^2 and PGS effect increases. Given significant and replicable evidence for context-specific PGS_{BMI} performance and effects, we investigated ways to increase model performance taking into account non-linear effects. Machine learning models (neural networks) increased relative model R^2 (mean 23%) across datasets. Finally, creating PGS_{BMI} directly from GxAge GWAS effects increased relative R^2 by 7.8%. These results demonstrate that certain covariates, especially those most associated with BMI, significantly affect both PGS_{BMI} performance and effects across diverse cohorts and

ancestries, and we provide avenues to improve model performance that consider these effects.

eLife assessment

This study presents a **valuable** analysis of the effects of covariates, such as age, sex, socio-economic status, or biomarker levels, on the predictive accuracy of polygenic scores for body mass index; it also presents approaches for improving prediction accuracy by accounting for such covariates. While the analyses are **solid**, the study falls short of providing a cogent interpretation of key findings, which could be of great interest and utility. The work will be of interest to people using and developing methods for phenotypic prediction based on polygenic scores.

Introduction

Polygenic scores (PGS) provide individualized genetic predictors of a phenotype by aggregating genetic effects across hundreds or thousands of loci, typically from genome-wide association studies (GWAS). In recent years it has become increasingly apparent that the transferability of PGS performance across different cohorts is poor (1). Most analyses to date have focused on ancestry differences as the main driver of this lack of transferability (2–4). However, a growing body of evidence has demonstrated that PGS performance and effect estimates are influenced by differences in certain environmental (classically termed "gene-environment" effects or interactions) or personal-level covariates – different phenotypes seem to be differently affected by these covariates, with most evidence being for adiposity traits such as body mass index (BMI) (5–14).

There are several gaps in current knowledge about these covariate-specific effects. Many analyses have assessed only a handful of these covariates due to the myriad of choices possible in typical large-scale biobanks. Little investigation has been done to systematically understand why certain covariates affect PGS performance, with such knowledge being useful to reduce the potential search for variables that impart context-specific effects. Furthermore, most studies investigating PGS-covariate interactions have been in European ancestry individuals; unfortunately, comparing differences in PGS performance and prediction while controlling for differences in ancestry versus differences in context has not been assessed in previous studies. Moreover, covariate-specific effects are notorious for having poor replicability in human genetics studies, and previous PGS-covariate interaction studies have been predominantly performed in the UK Biobank (UKBB) (15), where the majority of individuals are aged 40-69 (i.e., excluding young adults), are overall healthier than those from other e.g., hospital-based cohorts, and are predominantly European ancestry. Additionally, PGS performance is often assessed using linear models and in isolation of clinical covariates, which in practice would often be available. Machine learning models can have increased performance over linear models and are capable of modeling complex relationships and interactions between variables, which may serve to increase predictive performance, especially given evidence for PGS-covariate specific effects. Finally, given evidence for context-specific effects, it should be possible to directly incorporate SNP-covariate interaction effects from a GWAS directly to improve prediction performance, instead of relying on post-hoc interactions from a typical PGS calculated from main GWAS effects.

In this study we aimed to address the previously mentioned concerns. Using genetic data with linked-phenotypic information from electronic health records, we estimated the effects of covariate stratification and interaction on performance and effect estimates of PGS for BMI

(PGS_{BMI}) – **Figure 1** is a flowchart of the project. These analyses were done across four datasets: UK Biobank (UKBB), Penn Medicine BioBank (PMBB) (15), Electronic Medical Records and Genomics (eMERGE) network dataset (16), and Genetic Epidemiology Research on Adult Health and Aging (GERA). These datasets include participants from two ancestry groups (N=491,111 European ancestry (EUR), N=21,612 African ancestry (AFR)), and 62 covariates (25 present in multiple datasets) representing laboratory, survey, and biometric data types typically associated with cardiometabolic health and adiposity (including blood lipids, socioeconomic status, blood pressure, diet, physical activity, alcohol intake, smoking, and lung function, among others). We calculated the difference in PGS_{BMI} R^2 among individuals stratified by different quintiles of these variables (different groups for binary variables) and assessed whether a variable's association with BMI could explain its effect on PGS_{BMI} performance across groups. We also assessed the significance of PGS_{BMI}-covariate interaction terms, and their increases to model R^2 over models only using main effects, as well as correlation of main effect, interaction effect, and R^2 differences. Replication and comparisons across all datasets and ancestries were conducted. We then used machine learning models and evaluated their increase in performance over linear models, even those that included regularization and interaction terms. Finally, we conducted BMI GWAS using GxAge interaction terms, created PGS_{BMI} directly using these effects, and assessed performance gains over using typical main effect GWAS. This study addresses a plethora of open issues considering performance and effects of PGS on individuals from diverse backgrounds.

Results

Effect of covariate stratification on PGS_{BMI} performance

We assessed 62 covariates for R^2 differences (25 present, or suitable proxies, in multiple datasets) after stratifying on binary covariates and quintiles for continuous covariates. With UKBB EUR as discovery (N=376,729), 18 covariates had significant differences (Bonferroni p<.05/62) in R^2 among groups (**Figure 2a**), including age, sex, alcohol consumption, different physical activity measurements, Townsend deprivation index, different dietary measurements, lipids, blood pressure, and A1c, with 40 covariates having suggestive (p<.05) evidence of R^2 differences. From an original PGS_{BMI} R^2 of .076, R^2 increased to .094, .093, and .088 for those in the bottom physical activity, alcohol intake, and high-density lipoprotein (HDL) cholesterol quintiles, and decreased to .067, .57, and .049 for those in the top quintile (71%, 61%, and 56% relative R^2), respectively, comparable to differences due to ancestry (1). We note that the differences in R^2 due to alcohol intake and HDL were larger than those of any physical activity phenotype, despite physical activity having one of the oldest and most replicable evidence of interaction with genetic effects of BMI (33,34). Despite considerable published evidence suggesting covariate-specific genetic effects between BMI and smoking behaviors (6,8), we were only able to find suggestive evidence for R^2 differences when stratifying individuals across several smoking phenotypes (minimum p=0.016, for smoking pack years). R^2 differences due to educational attainment were also only suggestive (p=0.015), of which published evidence is conflicting (35– 37).

We replicated these analyses in three additional large-scale cohorts of European and African ancestry individuals (**Figure 2b**, Supplemental Table 3), as well as in African ancestry UKBB individuals. For covariates with significant performance differences in the discovery analysis, we were able to replicate significant (p<.05) R^2 differences for age, HDL cholesterol, alcohol intake frequency, physical activity, and HbA1c, despite much smaller sample sizes. We again observed largely insignificant differences across cohorts and ancestries when stratifying due to different smoking phenotypes and educational attainment. Combining across cohorts, for each covariate and ancestry we regressed R^2 values across groupings on covariate values in a weighted linear regression, weighting by sample size. Slope of a weighted linear regression across cohorts was

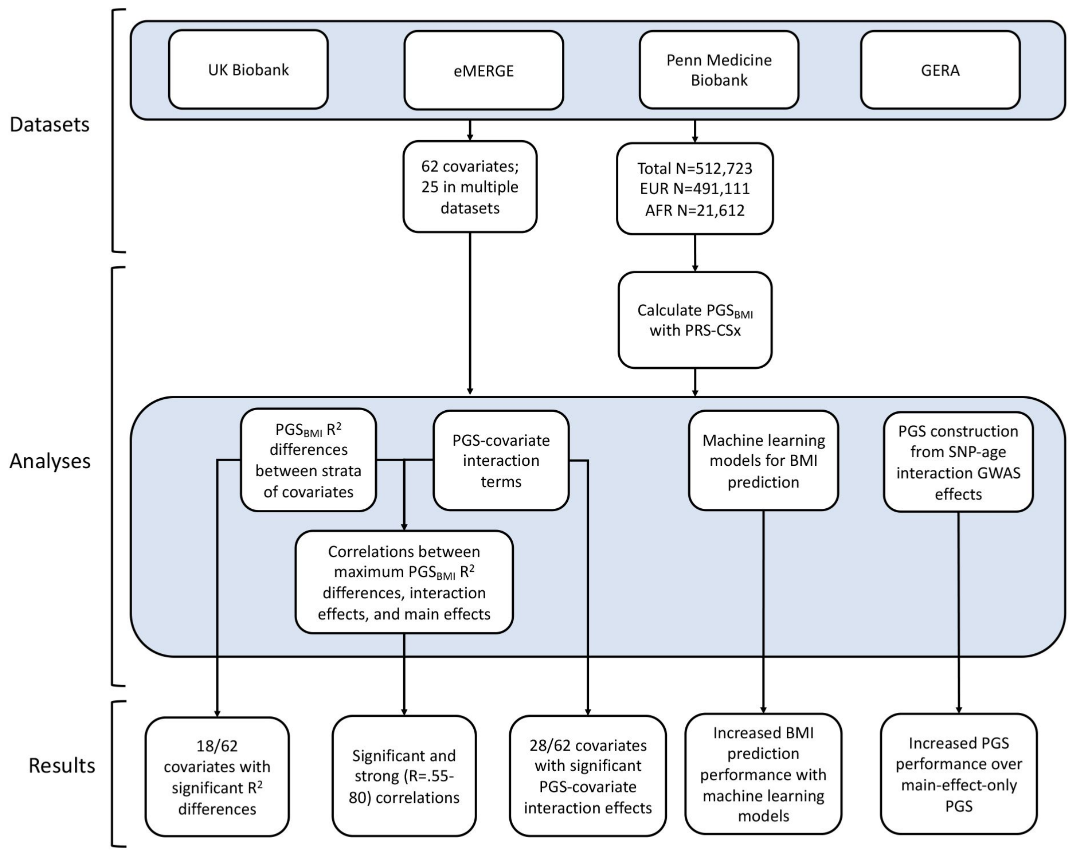

Figure 1.

A flowchart of the project.

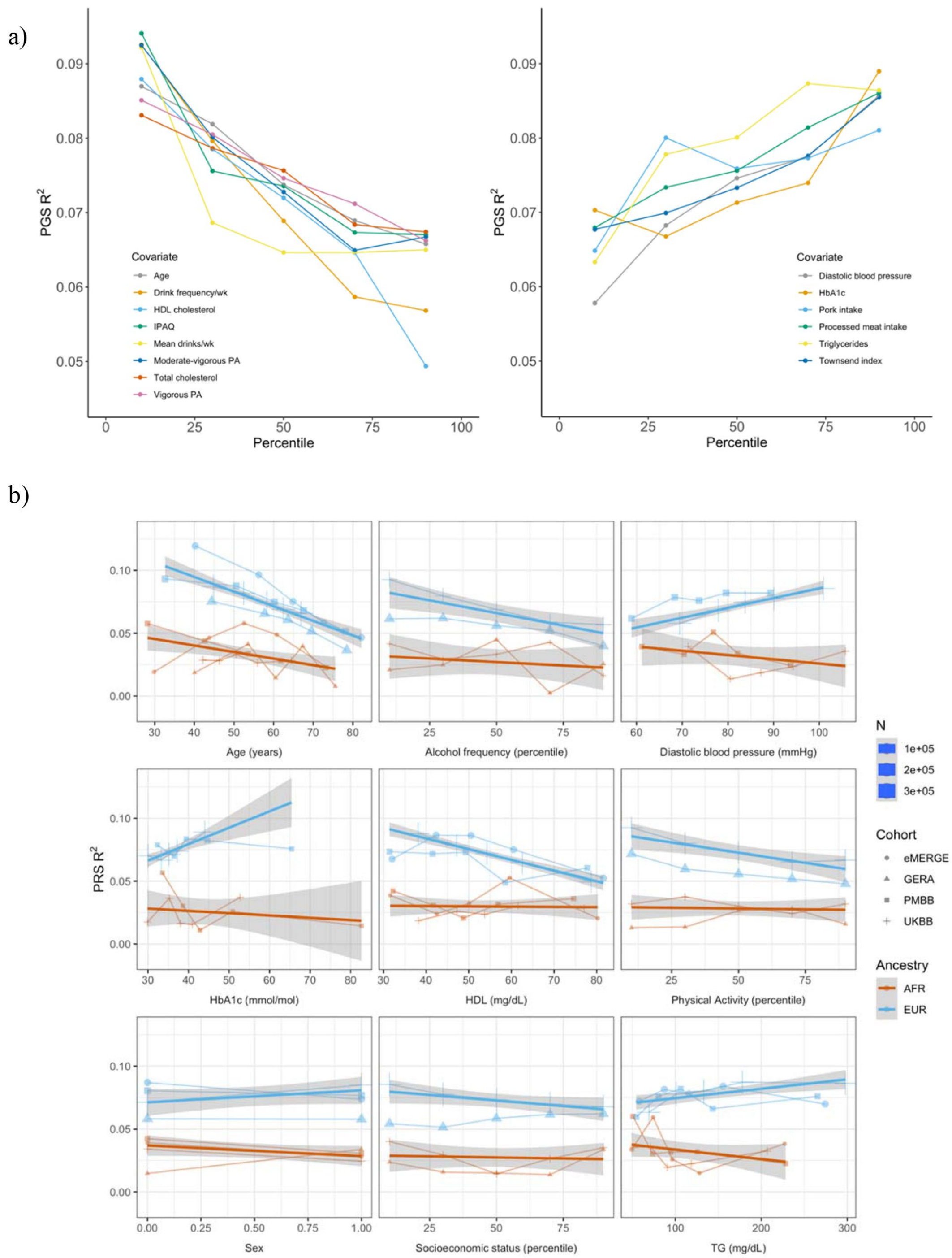

Figure 2.

PGS R^2 stratified by quintiles for quantitative variables and by binary variables. a) Continuous covariates with significant ($p < 7.7\times10^{-4}$) R^2 differences across quintiles in UKBB EUR. Pork and processed meat consumption per week were excluded from this plot in favor of pork and processed meat intake. b) Covariates with significant differences that were available in multiple cohorts, note actual values instead of percentiles plotted on x-axis (except percentiles were used for physical activity, alcohol intake frequency, and socioeconomic status, which had slightly differing phenotype definitions across cohorts). Townsend index and income were used as variables for socioeconomic status UKBB and GERA, respectively. Note that the sign for Townsend index was reversed, since increasing Townsend index is lower socioeconomic status, while increasing income is higher socioeconomic status. Abbreviations: physical activity (PA), International Physical Activity Questionnaire IPAQ).

directionally inconsistent between ancestries for the same covariate (triglyceride levels, HbA1c, diastolic blood pressure, and sex), although larger sample sizes may be needed to confirm these differences.

We noted several observations related to age-specific effects on PGS_{BMI}. First, in the weighted linear regression of all R^2 across ancestries, expected R^2 for African ancestry individuals overcomes that of European ancestry individuals among individuals within bottom and top age quintiles observed in these data. For instance, the predicted R^2 of .048 for 80 year-old European ancestry individuals would be lower than that of African ancestry individuals aged 24.7 and lower, indicating that certain differences in covariates can overcome differences in PGS_{BMI} performance due to ancestry (with age being one of the most common variables used for assessing and determining health). Second, we obtained these results despite the average age of GWAS individuals being 57.8, which should increase PGS_{BMI} R^2 of individuals closest to this age (13). This result suggests that in the case of these analyses, PGS performance due to decreased heritability with age cannot be reconciled by having GWAS individuals of similar age being used to create the PGS_{BMI}. Finally, we observed that PGS_{BMI} R^2 increases as age decreases, consistent with published evidence suggesting that the heritability of BMI decreases with age (38,39).

PGS-covariate interaction effects

Next, we estimated difference in PGS effects due to interactions with covariates, by modeling interaction terms between PGS_{BMI} for each covariate (Methods). We implemented a correction for shared heritability between covariates of interest and outcome (which can inflate test statistics (28)) to better measure the environmental component of each covariate, and show that this correction successfully reduces significance of interaction estimates (**Supplemental Figure 1**). Again, using UKBB EUR as discovery, we observed 28 covariates with significant (Bonferroni p<.05/62) PGS-covariate interactions (**Table 1**), with 38 having suggestive (p<.05) evidence (Supplemental Table 4). We observed the largest effect of PGS-covariate interaction with alcohol drinking frequency (20.0% decrease in PGS effect per 1 standard deviation (SD) increase, $p=2.62\times10^{-55}$), with large effects for different physical activity measures (9.4%-12.5% decrease/SD, minimum $p=3.11\times10^{-66}$), HDL cholesterol (15.3% decrease/SD, $p=1.71\times10^{-96}$) and total cholesterol (12.7% decrease/SD, $p=1.64\times10^{-71}$). We observed significant interactions with diastolic blood pressure (10.8% increase/SD, $p=6.06\times10^{-60}$), but interactions with systolic blood pressure were much smaller (1.17% increase/SD, $p=4.41\times10^{-3}$). Significant interactions with HbA1c (4.63% increase/SD, $p=5.37\times10^{-14}$) and type 2 diabetes (27.2% PGS effect increase in cases, $p=1.83\times10^{-7}$) were also observed. Other significant PGS-covariate interactions included lung function, age, sex, and LDL cholesterol – various dietary measurements also had significant interactions, albeit with smaller effects than other significant covariates. We were able to find significant interaction effects for smoking pack years (4.78% increase/SD, $p=3.68\times10^{-7}$), but other smoking phenotypes had insignificant interactions after multiple testing correction (minimum $p=2.7\times10^{-3}$) – interactions with educational attainment were also insignificant ($p= 4.54\times10^{-2}$).

We replicated these analyses across ancestries and the other cohorts in this study (**Figure 3**, Supplemental Table 4). For age and sex, which were available for all cohorts, interactions were significant (p<.05) and directionally consistent across all cohorts and ancestries (except for GERA AFR both effects were reversed, albeit with small sample size (N=1,789)). We were able to test interactions with alcohol intake frequency and physical activity in GERA, and we were able to replicate significant and directionally consistent association. We observed poor replication for LDL cholesterol, HbA1c, and smoking pack years, with insignificant and directionally inconsistent interaction effects across cohorts. Educational attainment was available in GERA, and interactions were once again insignificant. We observed significant and directionally consistent interaction effects for TG in eMERGE EUR and PMBB EUR, while the effect was inconsistent in UKBB EUR despite much larger sample size.

Variable type	Covariate	% change in β_{PGS} per covariate unit change	Interaction P	R^2 increase with interaction term	N
Continuous	HDL cholesterol	-15.29	1.71×10^{-96}	0.0012	328719
	Total cholesterol	-12.70	1.64×10^{-71}	0.00082	359221
	IPAQ	-12.50	3.11×10^{-66}	0.001	304951
	Moderate-vigorous PA	-11.41	8.92×10^{-65}	0.001	304951
	Diastolic BP	10.84	6.06×10^{-60}	0.0007	352804
	Townsend Index	6.78	2.86×10^{-58}	0.00089	376283
	Age	-9.02	3.60×10^{-57}	0.00061	376729
	FVC	-9.66	4.69×10^{-56}	0.0008	343467
	Drink frequency/wk	-19.96	2.62×10^{-55}	0.0024	122281
	LDL cholesterol	-9.86	2.63×10^{-51}	0.00058	358556
	N days vigorous PA/wk	-9.37	2.42×10^{-35}	0.0007	299963
	FEV1	-7.38	7.15×10^{-35}	0.0005	343544
	Mean alcohol consumption	-7.38	7.65×10^{-22}	0.00113	126756
	HbA1c	4.63	5.37×10^{-14}	0.0002	358798
	Mean drinks/wk	-7.66	1.01×10^{-13}	0.0008	112204
	Water intake	4.60	2.97×10^{-13}	0.00014	347472
	Processed meat intake	3.70	2.38×10^{-7}	0.0002	376205
	Starch mean	5.51	3.15×10^{-7}	0.00018	128346
	Smoking pack years	4.78	3.68×10^{-7}	0.0002	114135
	Protein mean	4.82	6.52×10^{-7}	0.00018	128181
	Saturated fat mean	4.92	1.23×10^{-6}	0.00017	127899
	Fat mean	4.40	1.64×10^{-5}	0.00013	128092
	Saturated fat grams/wk	2.46	1.79×10^{-5}	4.00E-05	364629
	Retinol mean	3.77	3.54×10^{-4}	9.00E-05	126029
Binary	IPAQ	-12.68	5.30×10^{-62}	0.0009	304951
	Vigorous PA/wk	-20.55	9.07×10^{-54}	0.0009	304951
	Sex	-11.02	1.41×10^{-24}	0.00025	376729
	Diabetes	27.19	1.83×10^{-7}	0.0004	375903

Table 1.

Model descriptive statistics on 28 of 62 covariates, which have significant ($p<.05/62$) PGS-covariate interaction terms, in UKBB EUR. The third column is the percentage change in PGS effect per unit change (standard deviations for continuous variables, binary variables encoded as 0 or 1) in covariate. The fifth column is the increase in model R^2 with a PGS-covariate interaction term versus a main effects only model. Abbreviations: blood pressure (BP), physical activity (PA), forced vital capacity (FVC), forced expiratory volume in 1-second (FEV1), International Physical Activity Questionnaire (IPAQ).

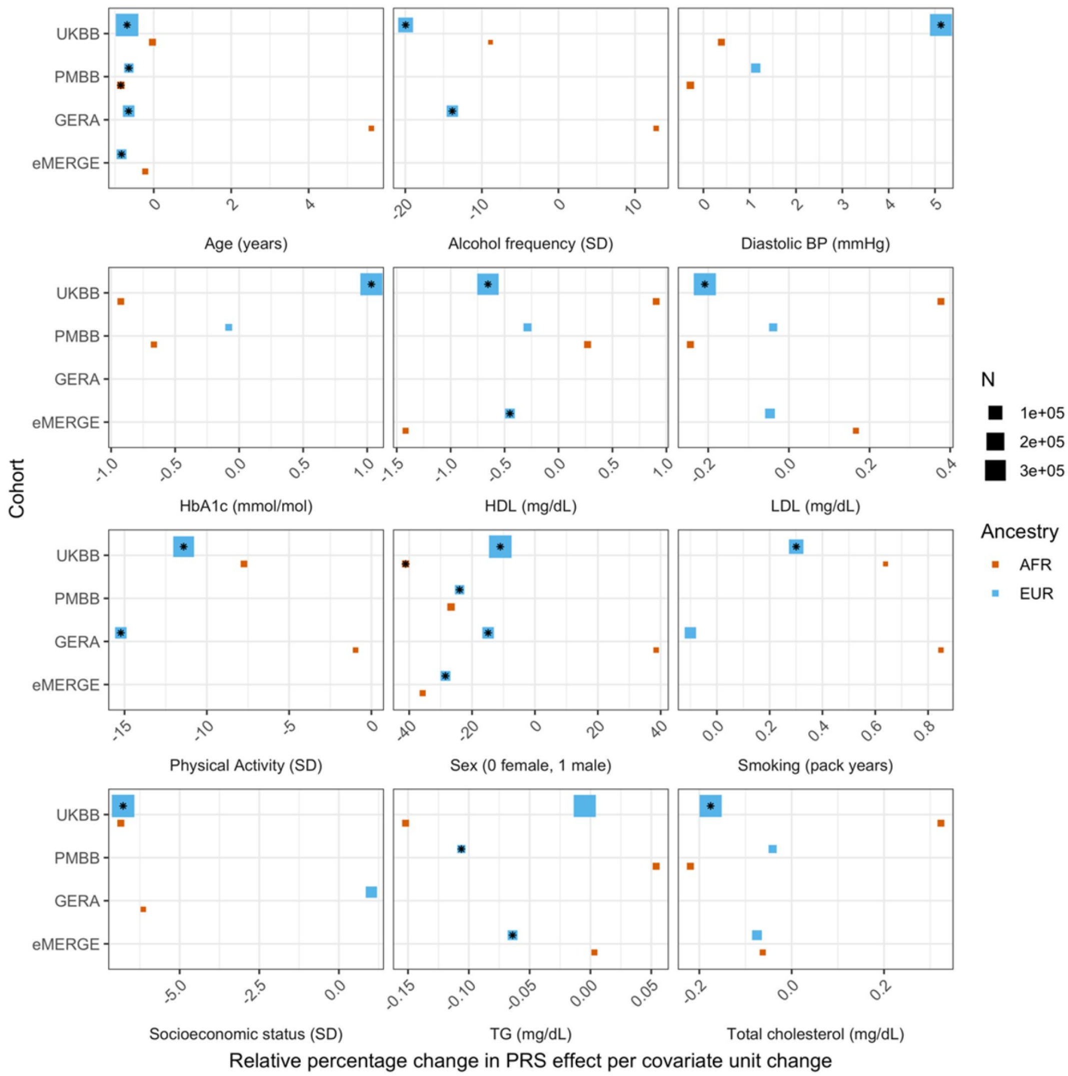

Figure 3.

Relative percentage changes in PGS effect per unit change in covariate, for covariates that significantly changed PGS effect (i.e., significant interaction beta at Bonferroni $p < 7.7\times10^{-4}$– denoted by asterisks) and were present in multiple cohorts and ancestries. Same covariate groupings and transformations were performed as with **Figure 1**.

However, despite significance of interaction terms, increases in model R^2 when including PGS-covariate interaction terms were small. For instance, the maximum increase among all covariates in UKBB EUR was only .0024 from a base R^2 of .1049 (2.1% relative increase), for alcohol drink frequency per week. Across all cohorts and ancestries, the maximum increase in R^2 was only .0058 from a base R^2 of 0.09454 (6.1% relative increase), when adding a PGS-age interaction term for eMERGE EUR ($p=5.40\times10^{-46}$) – this was also the largest relative increase among models with significant interaction terms. This result suggests that, while interaction effects can significantly modify the effect of PGS_{BMI}, their overall impact on model performance is relatively small, despite the observation that stratifying by covariates produces large differences in R^2.

Correlations between R^2 differences, interaction effects, and main effects

We next investigated the relationship between interaction effects, maximum R^2 differences across quintiles, and main effects of covariates on BMI. We first estimated main effects of each covariate on BMI (Methods, Supplemental Table 5), and then calculated the correlation weighted by sample size between main effects, maximum PGS_{BMI} R^2 across quintiles, and PGS-covariate interaction effects (**Figure 4**) across all cohorts and ancestries – GERA data were excluded from these analyses due to slightly different phenotype definition (Supplemental Table 6), as were binary variables. Interaction effects and maximum R^2 differences had a .80 correlation ($p=2.1\times10^{-27}$), indicating that variables with the largest interaction effects also had the largest effect on PGS_{BMI} performance across quintiles, and that covariates that increase PGS_{BMI} effect also have the largest effect on PGS_{BMI} performance i.e., individuals most at risk for obesity will have both disproportionately larger PGS_{BMI} effect and R^2. Main effects and maximum R^2 differences had a .56 correlation ($p=1.3\times10^{-11}$), while main effects and interaction effects had a .58 correlation (7.6×10^{-12}) again suggesting that PGS_{BMI} are more predictive in individuals with higher values of BMI-associated covariates, although predictive to a lower degree than estimating the interaction effects themselves directly. However, this result demonstrates that covariates that influence both PGS_{BMI} effect and performance can be predicted just using main effects of covariates, which are often known for certain phenotypes and easier to calculate, as genetic data and PGS construction would not be required.

Effects of machine learning approaches on predictive performance

Given evidence of PGS-covariate dependence, we aimed to assess increases in R^2 when using machine learning models (neural networks), which can inherently model interactions and other nonlinearities, over linear models even with interaction terms. We first included age and sex as the only covariates (along with genotype PCs and PGS_{BMI}), as age and sex were present in all datasets and had significant and replicable evidence for PGS-dependence across our analyses. Three models were assessed – L1-regularized (i.e., LASSO) linear regression without any interaction terms, LASSO including a PGS-age and PGS-sex interaction term, and neural networks (without interaction terms). When comparing neural networks to LASSO with interaction terms, the relative 10-fold cross-validated R^2 increased up to 67% (mean 23%) across cohorts and ancestries (**Figure 5**, Supplemental Table 7). The inclusion of interaction terms increased cross-validated R^2 up to 12% (mean 3.9%) when comparing LASSO including interaction terms to LASSO with main effects only.

We then modeled all available covariates for each cohort and did similar comparisons. Cross-validated R^2 increased up to 17.6% (mean 9.5%) when using neural networks versus LASSO with interaction terms, and up to 7.0% (mean 2.0%) when comparing LASSO with interaction terms to LASSO with main effects only. Increases in model performance using neural networks were smaller in UKBB, perhaps due to the age range being smaller than other cohorts (all participants aged 39-73). This result suggests that additional variance explained through non-linear effects with age and sex are explained by other variables present in the remainder of the datasets. Our

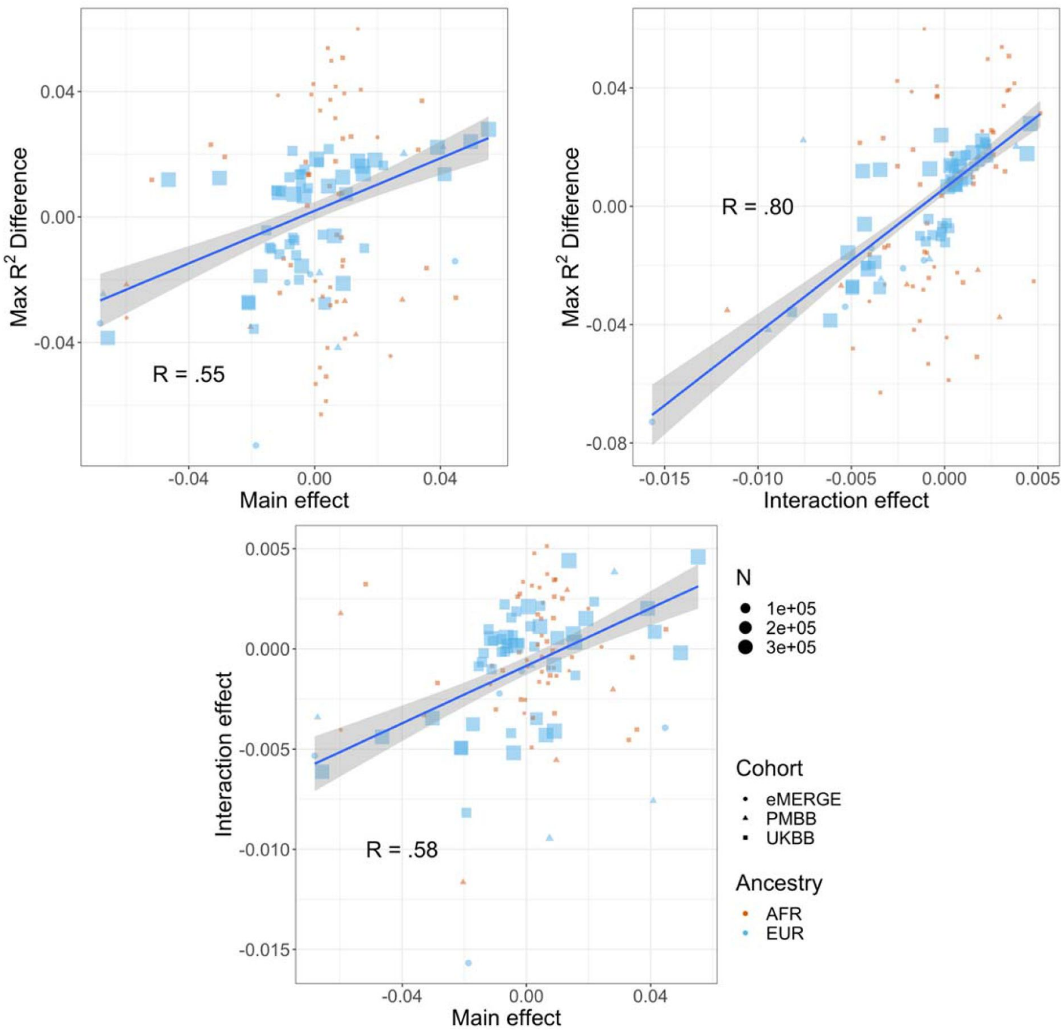

Figure 4.

Pearson correlations weighted by sample size between maximum R^2 differences across strata, main effects of covariate on log(BMI), and PGS-covariate interaction effects on log(BMI). Main effect units are in standard deviations, interaction effect units are in PGS standard deviations multiplied by covariate standard deviations. Only continuous variables are plotted and modeled. GERA was excluded due to slightly different phenotype definitions.

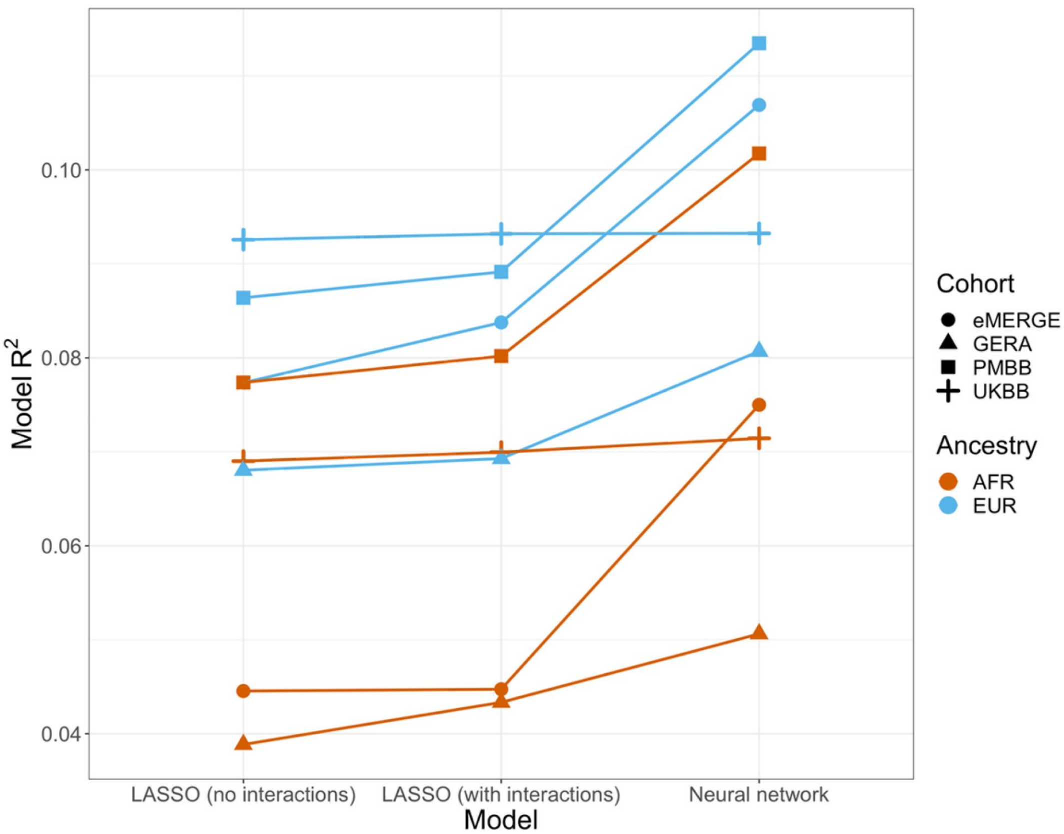

Figure 5.

Model R^2 across cohorts and ancestries using age and gender as covariates (along with PGS_{BMI} and PCs 1-5). Across all cohorts and ancestries, LASSO with PGS-age and PGS-gender interaction terms had better average 10-fold cross-validation R^2 than LASSO without interaction terms, while neural networks outperformed LASSO models.

findings suggest that machine learning methods can improve model R^2 that include PGS_{BMI} as variables beyond including interaction terms in linear models, even when variable selection is performed using LASSO.

Calculating PGS directly from GxAge GWAS effects

Previous studies (13) have created PGS using GWAS stratified by different personal-level covariates, but for practical purposes this leads to a large loss of power as the full size of the GWAS is not utilized for each strata and continuous variables are forced into bins. We developed a novel strategy where PGS are instead created from a full-size GWAS that includes SNP-covariate interaction terms (Methods). We focused on age interactions, given their large and replicable effects based on our results – similar to a previous study, we conducted these analyses in the European UKBB. We used a 60% random split of study individuals to conduct three sets of PGS using GWAS of the following designs: main effects only, main effects also with a SNP-age interaction term, and main effects but stratified into quartiles by age. 20% of the remaining data were used for training and the final 20% were held-out as a test set. The best performing PGS created from SNP-age interaction terms (PGS_{GxAge}) increased test R^2 to 0.0771 from 0.0715 (7.8% relative increase) compared to the best performing main effect PGS (**Figure 6**, Supplemental Table 9) – age-stratified PGS had much lower performance than both other strategies (unsurprising given reasons previously mentioned). Including a PGS_{GxAge}-age interaction term only marginally increased R^2 (.0001 increase), with similar increases for the other two sets of PGS, further demonstrating that post-hoc modeling of interactions cannot reconcile performance gained through directly incorporating interaction effects from the original GWAS. The strategy of creating PGS directly from full-sized SNP-covariate interactions is potentially quite useful as it increases PGS performance without the need for additional data – there are almost certainly a variety of points of improvement (Discussion), but we consider their investigation outside the scope of this study.

Discussion

Across four large-scale cohorts of diverse ancestry, we uncover replicable effects of covariates on both performance and effects of PGS_{BMI}. When stratifying by quintiles of different covariates, we find certain covariates that have significant and replicable evidence for differences in PGS_{BMI} R^2, with R^2 being nearly double between top and bottom performing quintiles for covariates with the largest differences. When testing covariates for PGS-covariate interaction effects, we also find covariates with significant interaction effects, where, for the largest effect covariates, each standard deviation change affects PGS_{BMI} effect by nearly 20%. Across analyses, we find age and sex had the most replicable interaction effects, with levels of serum cholesterol, physical activity, and alcohol consumption having the largest effects across cohorts. We also observed that interaction effects and R^2 differences were strongly correlated, with main effects also being correlated with interaction effects and R^2 differences, suggesting that covariates with the largest interaction effects also contribute to the largest R^2 differences, with simple main effects also being predictive of expected differences in R^2 and interaction effects. While in this study we stratified by quintile, similar trends of performance differences would be apparent in smaller percentiles. This observation is relevant to current research and clinical use of PGS, as individuals above a percentile cutoff are designated high risk (40). These analyses imply that individuals most at-risk for obesity should have both disproportionately higher predicted BMI and increased BMI prediction performance compared to the general population – more work is required to understand why, and if similar trends can be observed in other traits relevant to human health. We also employ machine learning methods for prediction of BMI with models that include PGS_{BMI} and demonstrate that these methods outperform regularized linear regression models that include

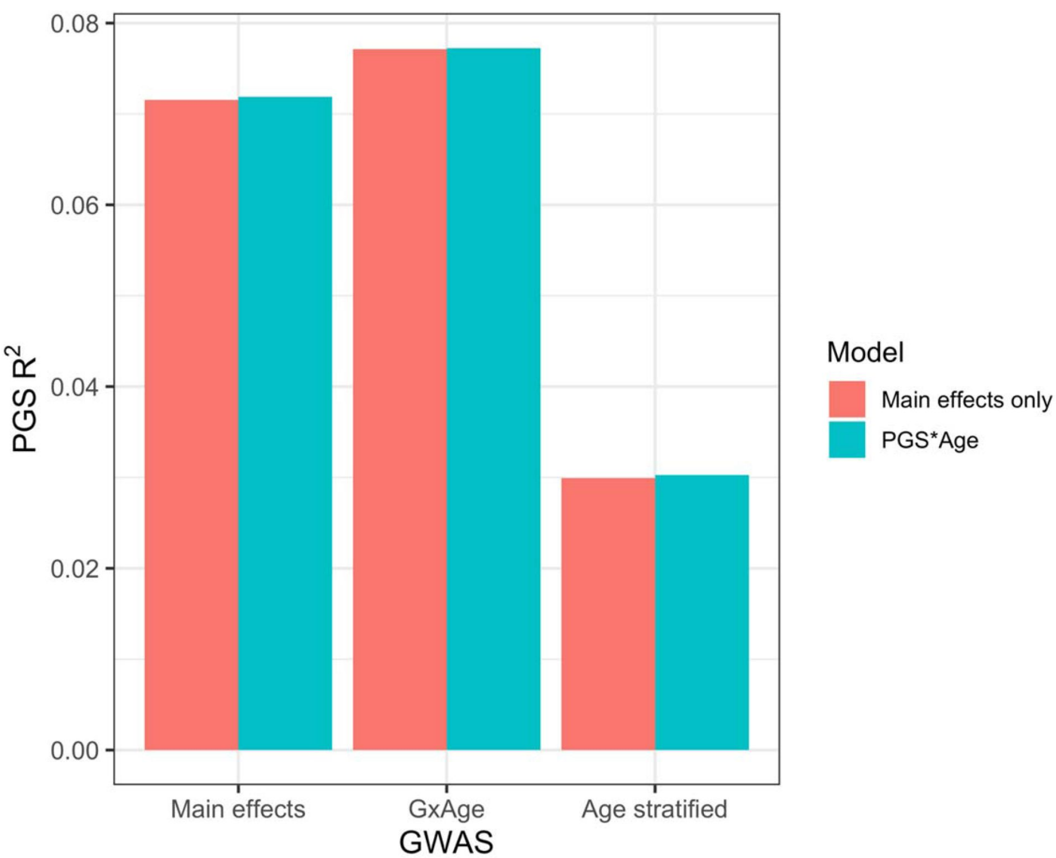

Figure 6.

PGS R^2 based on three sets of GWAS. “Main effects” were from a typical main effect GWAS, “GxAge” effects were from a GWAS with a SNP-age interaction term, and “Age stratified” GWAS had main effects only but were conducted in four age quartiles. PGS R^2 was evaluated using two models: one with main effects only, and one with an additional PGS*Age interaction term.

interaction effects. Finally, we employ a novel strategy that directly incorporates SNP interaction effects into PGS construction, and demonstrate that this strategy improves PGS performance when modeling SNP-age interactions compared to PGS created only from main effects.

Future work may include replicating these analyses across additional traits, and further exploring machine learning and deep learning methods on other phenotypes to determine if this trend of inclusion of PGS along with covariate interaction effects outperforms linear models for risk prediction. Additionally, the method of including a PGS for the covariate to better measure its environmental effect is potentially worth exploring further, which should improve in the future as PGS performance continues to increase. A slight limitation of this method in our study is that for the UKBB analyses the GWAS used for PGS construction were also from UKBB thus not out-of-sample, although many of the covariates only have GWAS available through UKBB individuals. Furthermore, a variety of improvements are likely possible when creating PGS directly from SNP-covariate interaction terms. First, we used the same SNPs that were selected by pruning and thresholding based on their main effect p-values, but selection of SNPs based on their interaction p-values should also be possible and would likely improve performance. Additionally, performance of pruning and thresholding-based strategies have largely been overtaken by methods that first adjust all SNP effects for LD and don't require exclusion of SNPs, and a method that could do a similar adjustment for interaction effects would likely outperform most current methods for traits with significant context-specific effects. Finally, incorporation of additional SNP-covariate interactions (e.g., SNP-gender) would also likely further improve prediction performance, although any SNP selection/adjustment procedures may further complicated by additional interaction terms.

We propose several ideas on why BMI-associated covariates have larger interaction effects and impact on R^2 for PGS_{BMI}. Age may be a proxy for accumulated gene-environment interactions as younger individuals have less exposure to environmental influences on weight compared to older individuals; therefore, one may expect that in younger individuals their phenotype could be better explained by genetics compared to environment. Secondly, PGS may more readily explain more extreme phenotypes because they incorporate large effect variants associated with very high weight or height (41). For BMI, the distribution is skewed toward the right i.e., effects in trait-increasing variants may have a larger potential to affect trait variation compared to trait-reducing variants. This explanation would likely be better suited to traits with left-tailed distributions and is not fully satisfactory as first log-transforming or rank-normal transforming the phenotype, as was done in this study, may invalidate this explanation.

PGS is a promising technique to stratify individuals for their risk of common, complex disease. In order to achieve accurate predictions as well as promote equity, further research is required regarding PGS methods and assessments. This research provides firm evidence on the context-specific nature of PGS and the incorporation of covariate interaction effects for predicting BMI and, apart from promoting equitable use of PGS across ancestries and cohorts, demonstrates ways to improve PGS performance incorporating context-specific effects.

Methods

Study datasets

Individual inclusion criteria and sample sizes per cohort are described below – additional information is available in Supplemental Table 1.

Ukbb

Individual-level quality-control and filtering have been described elsewhere (17) for European ancestry individuals. Briefly, individuals were split by ancestry according to both genetically inferred ancestry and self-reported ethnicity (15). Individuals with low genotyping quality and sex mismatch were removed, only unrelated individuals (pi-hat < .250) were retained, and variants were filtered at INFO > .30 and minor allele frequency (MAF) > .01. For African ancestry, individuals were first selected based on self-reported ethnicity “Black or Black British”, “Caribbean”, “African”, or “Any other Black background”. Individuals that were low quality i.e., “Outliers for heterozygosity or missing rate”, and that were Caucasian from “Genetic ethnic grouping” were removed. Of these individuals, those that were +/-6 standard deviations from the mean of the first 5 genetic principal components provided by UKBB were excluded. Finally, only unrelated individuals were retained up to the second degree using plink2 (18) “-king-cutoff .125”. After QC and consideration of phenotype, a total of 7,046 individuals in the UKBB AFR data who also had BMI available were used for downstream analyses. In total, 383,775 individuals were used for analysis (N_{EUR}=376,729, N_{AFR}=7,046).

eMERGE

Ancestry and relatedness inference have been described elsewhere (15). Individuals were split into European and African ancestry cohorts, and related individuals were removed (pi-hat > .250) such that all were unrelated. 35,064 individuals (N_{EUR}=31,961, N_{AFR}=3,103) were used for analysis.

Gera

Ancestry inference has been described elsewhere (19), and study individuals were divided into European and African ancestry cohorts. Related individuals were removed using plink2 “-king-cutoff .125”. 57,838 individuals (N_{EUR}=56,049, N_{AFR}=1,789) were used for analysis.

Pmbb

Ancestry inference and relatedness inference have been described elsewhere (20). Individuals were split into European and African ancestry cohorts, and related individuals were removed at pi-hat > .250. 36,046 individuals (N_{EUR}=26,372, N_{AFR}=9,674) were used for analysis.

Choice of covariates

A total of 62 covariates were included in the analyses, 25 of which were present (or similar proxies) in multiple datasets. These covariates were selected based on relevance to cardiometabolic health and obesity, and previous evidence of context-specific effects with BMI (5,6,8,21–23). For UKBB, phenotype values were used from the collection that was closest to recruitment, and for PMBB the median values were used – for GERA and eMERGE only one value was available. Additional details on covariate constructions, transformations, filtering, and cohort availability are in Supplemental Table 2.

PGS construction

PGS for BMI (PGS_{BMI}) were constructed using PRS-CSx (24), using GWAS summary statistics for individuals of European (25), African (26), and East Asian (27) ancestry that were out-of-sample of study participants. A set of 1.29 million HapMap3 SNPs provided by PRS-CSx was used for PGS calculation, which are generally well-imputed and variable frequency across global populations. Default settings for PRS-CSx (downloaded November 22, 2021) were used, which have been demonstrated to perform well for highly polygenic traits such as BMI (list of parameters in

Supplemental Table 8). The final PGS_{BMI} per ancestry and cohort was calculated by regressing log(BMI) on the PGS_{BMI} per ancestry without covariates – the combined, predicted value was used as a single PGS_{BMI} in downstream analyses.

For GERA, BMI was not transformed, as it was already binned into a categorical variable with 5 levels (18≤, 19-25, 26-29, 30-39, >40). Additionally, for GERA the uncombined ancestry-specific PGS_{BMI} were used in the final models, as it had higher R^2 than using the combined PGS_{BMI} (data not shown).

PGS_{BMI} performance after covariate stratification

Analyses were performed separately for each cohort and ancestry. For each covariate, individuals were binned by binary covariates or quintiles for continuous covariates. Incremental PGS_{BMI} R^2 was calculated by taking the difference in R^2 between:

$$\log(BMI) \sim PGS_{BMI} + Age + Sex + PCs_{1\text{-}5}$$
$$\log(BMI) \sim Age + Sex + PCs_{1\text{-}5}$$

We performed 5,000 bootstrap replications to obtain a bootstrapped distribution of R^2. P-values for differences in R^2 were calculated between groups by calculating the proportion of overlap between two normal distributions of the R^2 value using the standard deviations of the bootstrap distributions. Again for GERA, BMI was not transformed.

PGS_{BMI} interaction modelling

Evidence for interaction with each covariate with the PGS_{BMI} was evaluated using linear regression. It has been reported that the inclusion of covariates that share heritability with the outcome can inflate test statistic estimates (28–30). To assuage these concerns, we introduced a novel correction, where we first calculated a PGS for the covariate ($PGS_{Covariate}$) and included it in the model, as well as an interaction term between PGS_{BMI} and $PGS_{Covariate}$. The $PGS_{Covariate}$ terms were calculated using the European ancestry Neale Lab summary statistics (URLs) and PRS-CS (31). To standardize effect sizes across analyses, PGS_{BMI} and Covariate were first converted to mean zero and standard deviation of 1 (binary covariates were not standardized). We demonstrate inclusion of $PGS_{Covariate}$ terms successfully reduced significance of the PGS_{BMI}*Covariate term (**Supplemental Figure 1**). The final model used to evaluate PGS_{BMI} and Covariate interactions was:

$$\log(BMI) \sim PGS_{BMI}*Covariate + PGS_{BMI} + Covariate + PGS_{Covariate} + PGS_{BMI}*PGS_{Covariate} + Age + Sex + PCs_{1\text{-}5}$$

We report the effect estimates of the PGS_{BMI}*Covariate term, and differences in model R^2 with and without the PGS_{BMI}*Covariate term. Again for GERA, BMI was not transformed.

Correlation between R^2 differences, interaction effects, and main effects

We estimated main effects of each covariate on BMI with the following model:

$$\log(BMI) \sim Covariate + Age + Sex + PCs_{1\text{-}5}$$

Note that we ran new models with main effects only, instead of using the main effect from the interaction models (as the main effects in the interaction models depend on the interaction terms, and main effects used to create interaction terms are sensitive to centering of variables despite the scale invariance of linear regression itself (32)). We then estimated the correlation between main effects, interaction effects, and maximum R^2 differences across all cohorts and ancestries weighting by sample size, analyzing quantitative and binary variables separately.

Machine learning models

UKBB EUR and GERA EUR models were restricted to 30,000 random individuals, for computational reasons – BMI distributions did not differ from the full-sized datasets (Kolmogorov-Smirnov p-value of 0.29 and 0.57, respectively). PGS_{BMI} and top five genetic principal components were included as features in all models. Two sets of models were evaluated for each cohort and ancestry: including age and sex as features, and including all available covariates in each cohort as features. 'Ever Smoker' status was used in favor of 'Never' vs 'Current smoking' status (if present), as individuals with 'Never' vs 'Current' status are a subset of those with 'Ever Smoker' status. UKBB AFR with all covariates was excluded due to small sample size (N=53).

Neural networks were used as the model of choice, given their inherent ability model interactions and other nonlinear dependencies. Prior to modeling, all features were scaled to be between 0 and 1. We used average 10-fold cross validation R^2 to evaluate model performance. Separate models were trained using untransformed and log(BMI). L1-regularized linear regression was used with 18 values of lambda (1.0, 5.0×10^{-1}, 2.0×10^{-1}, 1.0×10^{-1}, 5.0×10^{-2}, 2.0×10^{-2}, ..., 1.0×10^{-5}, 5.0×10^{-6}, 2.0×10^{-6}). Models were trained without inclusion of interaction terms (which neural networks can implicitly model), using 1,000 iterations of random search with the following hyperparameter ranges: size of hidden layers [10, 200], learning rate [.01, .0001], type of learning rate [constant, inverse scaling], power t [.4, .6], momentum [.80, 1.0], batch size [32, 256], and number of hidden layers [1, 2].

GxAge PGS_{BMI} creation and assessment

Analyses were conducted in the European UKBB (N=376,629), as was done in a study on a similar topic (13). Three sets of analyses were performed, using GWAS conducted in a 60% random split of individuals using the following models (BMI was rank-normal transformed before GWAS):

1. BMI ~ SNP + Age + Sex + PCs_{1-5}
2. BMI ~ SNP + Age*SNP + Age + Sex + PCs_{1-5}
3. Using the model in 1) but stratified into quartiles by age. BMI was rank-normal transformed within each quartile.

Using each set of GWAS, PGS was first calculated in a 20% randomly selected training set of the dataset using pruning and thresholding using 10 p-value thresholds (.50, .10, ..., 5.0×10^{-5}, 1.0×10^{-5}) and remaining settings as default in Plink 1.9. For 2), GxAge PGS_{BMI} were calculated using SNPs clumped by their main effect p-values from 1), and additionally incorporating the GxAge interaction effects per SNP. In other words, instead of typical PGS construction as:

$$PGS_i = \beta_1 k_1 + \beta_2 k_2 + \ldots + \beta_n k_n$$

For an individual *i*'s PGS calculation, with main SNP effect β, and *n* SNPs, PGS incorporating GxAge effects (PGS_{GxAge}) was calculated as:

$$PGS_{GxAge,i} = \beta_1 k_1 + \beta_{GxAge,1} k_1 Age_i + \beta_2 k_2 + \beta_{GxAge,2} k_2 Age_i \ldots + \beta_n k_n + \beta_{GxAge,n} k_n Age_i$$

Where β_{GxAge} is the GxAge effect for each SNP *n* and Age_i is the age for individual *i*.

For each of the three analyses, the parameters and models resulting in the best performing PGS (highest incremental R^2, using same main effect covariates as in the three GWAS) from the training set were evaluated in the remaining 20% of the study individuals. For 3), models were first trained within each quartile separately. To maintain sense of scale across quartiles (after rank-normal transformation), R^2 between all predicted values and true values was calculated together.

URLs

Neale Lab UKBB summary statistics: *http://www.nealelab.is/uk-biobank*

Data Availability statement

UK Biobank data was accessed under project #32133. eMERGE data is available at dbGaP in phs001584.v2.p2. GERA data is available at dbGaP in phs000674.v3.p3. Summary statistics for PMBB are available from authors upon request.

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Data Availability

UK Biobank data was accessed under project #32133. eMERGE data is available at dbGaP in phs001584.v2.p2. GERA data is available at dbGaP in phs000674.v3.p3. Summary statistics for PMBB are available from authors upon request.

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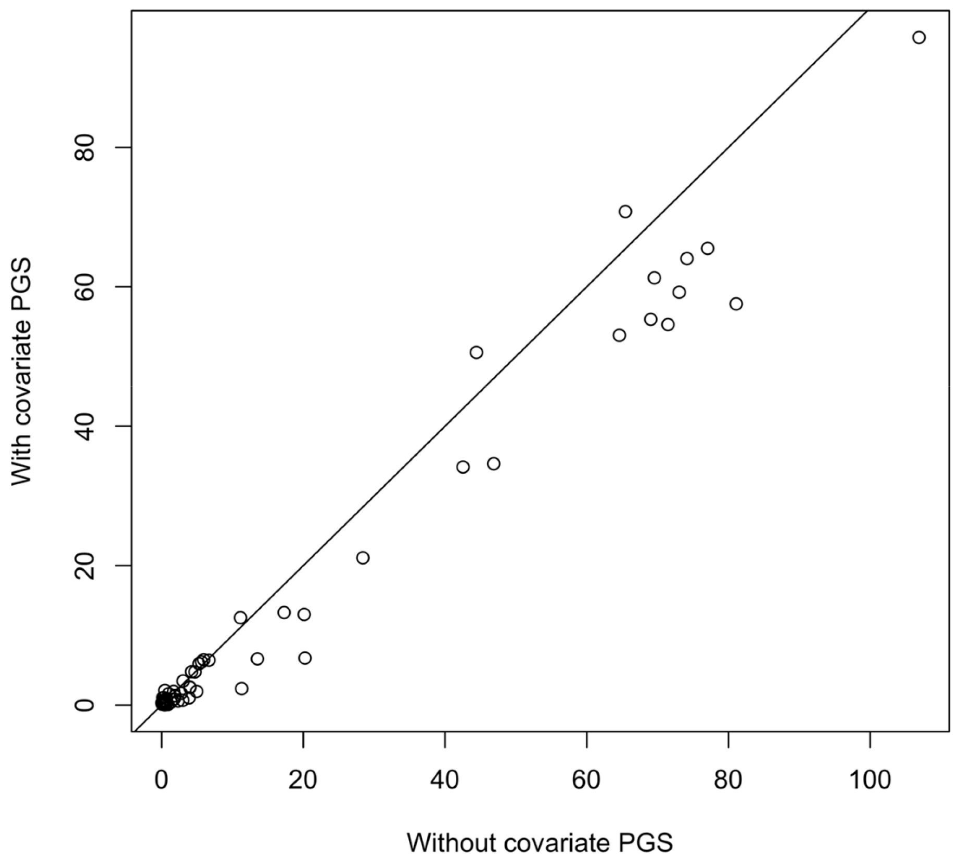
Supplemental Figure



S Figure 1.

PGS-covariate interaction term p-values in UKBB EUR, with and without including the covariate PGS in the model – the mean $-\log_{10}(p)$ is reduced from 18.0899 to 14.97072 with their inclusions. Note age and sex PGS were not calculated, and their interaction p-values are excluded from this figure.

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Reviewer #1 (Public Review):

In this paper, Hui and colleagues investigate how the predictive accuracy of a polygenic score (PGS) for body mass index (BMI) changes when individuals are stratified by 62 different covariates. After showing that the PGS has different predictive power across strata for 18 out of 62 covariates, they turn to understanding why these differences and seeing if predictive performance could be improved. First, they investigated which types of covariates result in the largest differences in PGS predictive power, finding that covariates with larger "main effects" on the trait and covariates with larger interaction effects (interacting with the PGS to affect the trait) tend to better stratify individuals by PGS performance. The authors then see if including interactions between the PGS and covariates improves predictive accuracy, finding that linear models only result in modest increases in performance but nonlinear models result in more substantial performance gains.

Overall, the results are interesting and well-supported. The results will be broadly interesting to people using and developing PGS methods. Below I list some strengths and minor weaknesses.

Strengths:

A major impediment to the clinical use of PGS is the interaction between the PGS and various other routinely measured covariates, and this work provides a very interesting empirical study along these lines. The problem is interesting, and the work presented here is a convincing empirical study of the problem.

The result that PGS accuracy differs across covariates, but in a way that is not well-captured by linear models with interactions is important for PGS method development.

Weakness:

While arguably outside the scope of this paper, one shortcoming is the lack of a conceptual model explaining the results. It is interesting and empirically useful that PGS prediction accuracy differs across many covariates, but some of the results are hard to reconcile simultaneously. For example, it is interesting that triglyceride levels are associated with PGS performance across cohorts, but it seems like the effect on performance is discordant across datasets (Figure 2). Similarly, many of these effects have discordant (linear) interactions across cohorts (Figure 3). Overall it is surprising that the same covariates would be important but for presumably different reasons in different cohorts. Similarly, it would be good to discuss how the present results relate to the conceptual models in Mostafavi et al. (eLife 2020) and Zhu et al. (Cell Genomics 2023).

Reviewer #2 (Public Review):

This work follows in the footsteps of earlier work showing that BMI prediction accuracy can vary dramatically by context, even within a relatively ancestrally homogenous sample. This is an important observation that is worth the extension to different context variables and samples.

Much of the follow-up analyses are commendably trying to take us a step further-towards explaining the underlying observed trends of variable prediction accuracy for BMI. Some of these analyses, however, are somewhat confounded and hard to interpret.

For example, many of the covariates which the authors use to stratify the sample by may drive range restriction effects. Further, the covariates considered could be causally affected by genotype and causally affect BMI, with reverse causality effects; other covariates may be partially causally affected by both genotype and BMI, resulting in collider bias. Finally, population structure differences between quintiles of a covariate may drive variable levels of stratification. These can bias estimation and confounds interpretations, at least one of which intuitively seems like a concern for each of the context variables (e.g., the covariates SES, LDL, diet, age, smoking, and alcohol drinking).

The increased prediction accuracy observed with some of the age-dependent prediction models is notable. Despite the clear utility of this investigation, I am not aware of much existing work that shows such improvements for context-aware prediction models (compared to additive/main effect models). I would be curious to see if the predictive utility extends to held-out data from a data set distinct from the UKB, where the model was trained, or whether it replicates when predicting variation within families. Such analyses could strengthen the evidence for these models capturing direct causal effects, rather than other reasons for the associations existing in the UKB sample.

Reviewer #3 (Public Review):

Polygenic scores (PGS), constructed based on genetic effect sizes estimated in genome-wide association studies (GWAS) and used to predict phenotypes in humans have attracted considerable recent interest in human and evolutionary genetics, and in the social sciences. Recent work, however, has shown that PGSs have limited portability across ancestry groups, and that even within an ancestry group, their predictive accuracy varies markedly depending on characteristics such as the socio-economic status, age, and sex of the individuals in the samples used to construct them and to which they are applied. This study takes further steps in investigating and addressing the later problem, focusing on body mass index, a phenotype of substantial biomedical interest. Specifically, it quantifies the effects of a large number of co-variates and of interactions between these covariates and the PGS on prediction accuracy; it also examines the utility of including such covariates and interaction in the construction of predictors using both standard methods and artificial neural networks. This study would be of interest to investigators that develop and apply PGSs.

I should add that I have not worked on PGSs and am not a statistician, and apologize in advance if this has led to some misunderstandings.

Strengths:

- The paper presents a much more comprehensive assessment of the effects of covariates than previous studies. It finds many covariates to have a substantial effect, which further highlights the importance of this problem to the development and application of PGSs for BMI and more generally.
- The findings re the relationships between the effects of covariates and interactions between covariates and PGSs are, to the best of my knowledge, novel and interesting.
- The development of predictors that account for multiple covariates and their interaction with the PGS are, to the best of my knowledge, novel and may prove useful in future efforts to produce reliable PGSs.
- The improvement offered by the predictors that account for PGS and covariates using neural networks highlights the importance of non-linear interactions that are not addressed by standard methods, which is both interesting and likely to be of future utility.

Weaknesses:

- The paper would benefit substantially from extensive editing. It also uses terminology that is specific to recent literature on PGSs, thus limiting accessibility to a broader readership.
- The potential meaning of most of the results is not explored. Some examples are provided below:
• the paper emphasizes that 18/62 covariates examined show significant effects, but this result clearly depends on the covariates included. It would be helpful to provide more detail on how these covariates were chosen. Moreover, many of these covariates are likely to be correlated, making this result more difficult to interpret. Could these questions at least be partially addressed using the predictors constructed using all covariates and their interactions jointly (i.e., with LASSO)? In that regard, it would be helpful to know how many of the covariates and interactions were used in this predictor (I apologize if I missed that).
• While the relationship between covariate effects and covariate-PGS interaction effects is intriguing, it is difficult to interpret without articulating what one would expect, i.e., what would be an appropriate null.
• The finding that using artificial neural networks substantially improves prediction over more standard methods is especially intriguing, and highlights the potential importance of non-linear relationships between PGSs and covariates. These relationships remain hidden in a black box, however. Even fairly straightforward analyses, based on using different combinations of the PGS and/or covariates may shed some light on these relationships. For example, analyzing which covariates have a substantial effect on the prediction or varying one covariate at a time for different values of the PGS, etc.
- The relationship to previous work should be discussed in greater detail.

Author Response:

Reviewer #1 (Public Review):

> *In this paper, Hui and colleagues investigate how the predictive accuracy of a polygenic score (PGS) for body mass index (BMI) changes when individuals are stratified by 62 different covariates. After showing that the PGS has different predictive power across strata for 18 out of 62 covariates, they turn to understanding why these differences and seeing if predictive performance could be improved. First, they investigated which types of covariates result in the largest differences in PGS predictive power, finding that covariates with larger "main effects" on the trait and covariates with larger interaction effects (interacting with the PGS to affect the trait) tend to better stratify individuals by PGS performance. The authors then see if including interactions between the PGS and covariates improves predictive accuracy, finding that linear models only result in modest increases in performance but nonlinear models result in more substantial performance gains.*
>
> *Overall, the results are interesting and well-supported. The results will be broadly interesting to people using and developing PGS methods. Below I list some strengths and minor weaknesses.*
>
> *Strengths:*
>
> *A major impediment to the clinical use of PGS is the interaction between the PGS and various other routinely measured covariates, and this work provides a very interesting empirical study along these lines. The problem is interesting, and the work presented here is a convincing empirical study of the problem.*
>
> *The result that PGS accuracy differs across covariates, but in a way that is not well-captured by linear models with interactions is important for PGS method development.*

Thank you for all of the positive comments.

> *Weakness:*
>
> *While arguably outside the scope of this paper, one shortcoming is the lack of a conceptual model explaining the results. It is interesting and empirically useful that PGS prediction accuracy differs across many covariates, but some of the results are hard to reconcile simultaneously. For example, it is interesting that triglyceride levels are associated with PGS performance across cohorts, but it seems like the effect on performance is discordant across datasets (Figure 2). Similarly, many of these effects have discordant (linear) interactions across cohorts (Figure 3). Overall it is surprising that the same covariates would be important but for presumably different reasons in different cohorts. Similarly, it would be good to discuss how the present results relate to the conceptual models in Mostafavi et al. (eLife 2020) and Zhu et al. (Cell Genomics 2023).*

Thank you for the comments. We agree that more generalizable explanations would be useful, which may be worth exploring in future work. Specifically, if there is heteroskedasticity in the relationship between PGS and BMI (e.g., phenotypic variance increases for higher values of BMI while PGS variance does not, or at least by a different amount), then that may partially explain the performance differences when stratifying by covariates that have main effects on BMI – somewhat similarly to what is presented in Figure

2 of Mostafavi et al. Such results may imply that similar performance differences could occur when stratifying by the phenotype itself, although this still may not explain differences in PGS effects, and differences in performance when using nonlinear methods (such as in this work and in Figure 4 of Zhu et al.). While we observe discordant effects for certain covariates across datasets, the findings from the correlation analyses use all cohorts and ancestries, and we expect that these difference in effects across datasets may be due to differences in their relationship with BMI across datasets (triglyceride levels may be especially noisy due to their sensitivity to fasting which may have been controlled for differently across datasets).

Reviewer #2 (Public Review):

> *This work follows in the footsteps of earlier work showing that BMI prediction accuracy can vary dramatically by context, even within a relatively ancestrally homogenous sample. This is an important observation that is worth the extension to different context variables and samples.*
>
> *Much of the follow-up analyses are commendably trying to take us a step further- towards explaining the underlying observed trends of variable prediction accuracy for BMI. Some of these analyses, however, are somewhat confounded and hard to interpret.*
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> *For example, many of the covariates which the authors use to stratify the sample by may drive range restriction effects. Further, the covariates considered could be causally affected by genotype and causally affect BMI, with reverse causality effects; other covariates may be partially causally affected by both genotype and BMI, resulting in collider bias. Finally, population structure differences between quintiles of a covariate may drive variable levels of stratification. These can bias estimation and confounds interpretations, at least one of which intuitively seems like a concern for each of the context variables (e.g., the covariates SES, LDL, diet, age, smoking, and alcohol drinking).*
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Thank you for the comments. We agree there are certain biases that may be introduced in these analyses. For population structure between quintiles, the analyses are already stratified by ancestry and have the top 5 genetic principal components included, which may help with this issue. In the interaction models we included separate terms for the PGS of the covariate as well which was meant to better capture the environmental component of the covariates, which may partially ameliorate issues of collider bias as SNPs that are causally affecting both BMI and the covariate would be partially adjusted for. While range restriction effects could introduce bias, in the correlation analyses the relationship between main effects and interaction effects (which were estimated without range restriction) have strong and reproducible correlations with PGS R2 differences across datasets.

We agree the increased prediction performance using PGS created directly from GxAge GWAS effects is notable, as it is essentially “free” performance increase that doesn’t require any new data, and it likely generalizable to additional covariates. It would be useful to validate its performance in other datasets, especially ones that are outside of the 40-69 age of UKBB.

Reviewer #3 (Public Review):

> *Polygenic scores (PGS), constructed based on genetic effect sizes estimated in genome-wide association studies (GWAS) and used to predict phenotypes in humans have attracted considerable recent interest in human and evolutionary genetics, and in the social sciences. Recent work, however, has shown that PGSs have limited portability across ancestry groups, and that even within an ancestry group, their predictive accuracy varies markedly depending on characteristics such as the socio-economic status, age, and sex of the individuals in the samples used to construct them and to which they are applied. This study takes further steps in investigating and addressing the later problem, focusing on body mass index, a phenotype of substantial biomedical interest. Specifically, it quantifies the effects of a large number of co-variates and of interactions between these covariates and the PGS on prediction accuracy; it also examines the utility of including such covariates and interaction in the construction of predictors using both standard methods and artificial neural networks. This study would be of interest to investigators that develop and apply PGSs.*
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Thank for the positive feedback.

> *Weaknesses:*
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> - *The paper would benefit substantially from extensive editing. It also uses terminology that is specific to recent literature on PGSs, thus limiting accessibility to a broader readership.*
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interaction effects is intriguing, it is difficult to interpret without articulating what one would expect, i.e., what would be an appropriate null. • The finding that using artificial neural networks substantially improves prediction over more standard methods is especially intriguing, and highlights the potential importance of non-linear relationships between PGSs and covariates. These relationships remain hidden in a black box, however. Even fairly straightforward analyses, based on using different combinations of the PGS and/or covariates may shed some light on these relationships. For example, analyzing which covariates have a substantial effect on the prediction or varying one covariate at a time for different values of the PGS, etc.

- *The relationship to previous work should be discussed in greater detail.*

Thank you for the comments. Regarding running LASSO with all covariates along with each of their interactions with PGS in one model, upon reading those sections of the text again it is a little unclear we agree; but we actually did something very similar already (related sections have been edited for clarity in our revised manuscript) with these results being presented later on in the neural network section (second paragraph, S Table 7 – those results specifically aren't in Figure 5). We just looked at changes in prediction performance, and did not try to interpret the model coefficients. We agree that many of the covariates are probably correlated, but based on the correlation results (Figure 4) it doesn't seem like any covariate is especially important separately from its effect on BMI itself i.e., whatever covariates were chosen by LASSO may still not be especially important. This explanation is related to the interpretation of the neural network results, where neural networks improved performance even over linear models with just age and sex and their interactions with PGS as additional covariates, which may suggest that increased performance is coming from nonlinearities apart from multiplicative interaction effects with the PGS. So observing the coefficients from LASSO but still with a linear model may still not substantially aid in explaining the relationships that increase prediction performance using neural networks (additionally, this analysis may be difficult to replicate since many of the covariates are not present in multiple datasets). But this replication would be nice to see in future studies if such datasets exist. In terms of the null relationship between covariate main and interaction effects, if they are from the same model they will inherently be correlated, but the main effects from Figure 4 are from a main effects model only. Regarding the other points, the text will be edited for clarity and elaboration on specific topics.